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ASSIGNMENT-2

Real-World Application: Online Video Streaming Services

1. Parallel Computing in video streaming services.

- When you upload or stream video, services like NETFLIX transcode that video into multiple resolutions and formats.
- Parallel computing is used to split and process video files across multiple processors at once.

It is important

- Reduce wait time for users to upload and access videos in all needed formats.
- Helps the service scale to millions of users or new content daily without performance delays.

2. Networked Systems in video Streaming

- Content is stored and delivered via a globally distributed network of servers, called a Content Delivery Network.
- When a user presses play, the video is streamed from the nearest CDN server, minimizing latency.
- User request, playback data, and performance metrics are consistently transmitted.

Why It's Important!

- Ensures fast, smooth video playback regardless of location.
- Supports millions of concurrent users by balancing load across global servers.